

README / METHODOLOGY

Project Title

Online Retail Business Analysis

Dataset

- Dataset: UCI Online Retail II
- Source: UCI Machine Learning Repository
- Data Period: 2009–2011
- Dataset contains more than 1 million transaction records.

Objective

The objective of this analysis is to understand transaction patterns, product demand, market concentration, and possible business improvement opportunities.

Methodology

1. Data Collection
2. Data Cleaning
3. Data Processing
4. Exploratory Data Analysis
5. Business Insights
6. Recommendations

Data Cleaning

- Missing Customer IDs were identified and retained because other transaction information was still useful.
- Duplicate records were checked and 6,865 duplicate rows were removed.
- Negative quantity values were identified and retained because they may represent returns or cancellations.
- Invoice date format was checked.

Analysis Performed

- Daily transaction quantity analysis
- Invoice/transaction frequency analysis
- Country-wise analysis
- Product-wise quantity analysis
- Comparison of transaction volume and transaction frequency

Key Findings

- Highest daily quantity observed was 87,167 units.
- Highest invoice activity observed was 675 invoices.
- The United Kingdom had the highest transaction volume.
- Several products showed high transaction quantities.
- Transaction volume and transaction frequency showed different patterns.

Recommendations

- Improve inventory planning for high-demand products.
- Increase operational capacity during peak transaction periods.
- Explore growth opportunities in other European markets.

Limitations

- Some Customer ID values were missing.
- Negative quantities may represent returns or cancellations.
- Analysis was mainly based on quantity and invoice count.
- The analysis does not establish causal relationships.

Conclusion

The analysis highlights important transaction, product, and market patterns that can support better inventory planning, operational planning, and market expansion decisions.